East West University

Course Title: Software Engineering (CSE412)

Section: 02

Project: School Management Software

Submitted by –

Group:04

Student Name	Student Id	Role
Kowser Ahmed	2023-3-60-144	Team Lead
MD.Minhajul Hoque Khandoker	2023-3-60-060	
Md. Maidul Islam	2023-3-60-068	
Moomtahina Zahin	2023-2-60-190	

Submitted to –

Dr. Mohammad Mahdi Hassan

Associate Professor

Department of Computer Science & Engineering

Project Report Update

Project Title

School Management Software

1. Introduction

The School Management Software with AI Integration is designed to automate and modernize the daily operations of educational institutions. The proposed system replaces traditional manual processes with a centralized digital platform that efficiently manages students, teachers, attendance, examinations, fees, and communication. Artificial Intelligence is integrated to provide smart features such as performance analytics, AI chatbot support, attendance anomaly detection, automated report generation, and intelligent notifications.

2. Problem Statement

Many schools still rely on manual record-keeping systems for attendance, examinations, fee management, and communication. These methods are inefficient, time-consuming, and prone to errors. As the number of students and staff increases, managing information manually becomes increasingly difficult. Therefore, a centralized software solution is required to improve efficiency, accuracy, and decision-making.

3. Project Objectives

The main objectives of the project are:

- Develop a centralized school management platform.
- Automate attendance, examination, and fee management.
- Implement secure role-based authentication.
- Integrate AI-powered analytics and chatbot services.
- Improve communication using SMS and email notifications.
- Provide an easy-to-use interface for administrators, teachers, students, parents, and accountants.

4. Stakeholders

- School Administrator
- Teachers
- Students
- Parents
- Accountant
- System Administrator

5. Functional Requirements

The system shall provide the following functionalities:

- User Authentication
- Student Management
- Teacher & Staff Management
- Attendance Management
- Examination & Result Management
- Routine Management
- Accounts & Fee Management
- Salary & Payroll Management
- Admission Management
- SMS & Email Notification
- AI Chatbot
- AI Performance Analytics
- AI Report Generator
- Website CMS
- Support Ticket System

6. Non-Functional Requirements

- Secure Authentication using JWT
- Role-Based Access Control (RBAC)
- Responsive User Interface
- High System Availability
- Fast Response Time
- Database Backup and Recovery
- Audit Logging
- Scalability
- Maintainability

7. User Stories

User Stories, Priorities & Acceptance Criteria

User Story 01 (Priority: High)

As an Administrator,

I want to manage student information,

So that I can maintain accurate student records.

Acceptance Criteria

- Admin can add a new student.
- Admin can update student information.
- Admin can delete inactive student records.
- Student ID must be unique.
- The system displays a success message after each operation.

User Story 02 (Priority: High)

As a Teacher,

I want to record student attendance,

So that attendance records remain accurate and up to date.

Acceptance Criteria

- Teacher can mark attendance once per class.
- Attendance is stored immediately.
- Duplicate attendance entries are prevented.
- Teacher can edit attendance before the class closes.
- Attendance reports can be generated.

User Story 03 (Priority: High)

As a Student,

I want to view my examination results,

So that I can monitor my academic performance.

Acceptance Criteria

- Student can view published results.
- GPA and grades are displayed correctly.
- Students cannot modify results.
- Results are available by semester/exam.
- The system loads results within a few seconds.

User Story 04 (Priority: High)

As an Accountant,

I want to manage student fee payments,

So that financial records remain accurate.

Acceptance Criteria

- Accountant can record fee payments.
- Outstanding dues are calculated automatically.
- Payment receipts can be generated.
- Payment history is available.
- Duplicate transactions are prevented.

User Story 05 (Priority: Medium)

As a Parent,

I want to receive attendance and result notifications,

So that I stay informed about my child's academic progress.

Acceptance Criteria

- Parents receive SMS or email notifications.
- Notifications are sent after attendance updates.
- Result publication triggers a notification.
- Parents can view previous notifications.
- Invalid contact information is handled gracefully.

User Story 06 (Priority: High)

As an Administrator,

I want to manage teacher and staff information,

So that employee records remain organized.

Acceptance Criteria

- Admin can add teachers and staff.
- Admin can update employee information.
- Admin can assign departments and subjects.
- Employee IDs must be unique.
- Records can be searched quickly.

User Story 07 (Priority: Medium)

As a Student,

I want to interact with an AI chatbot,

So that I can quickly get answers to common academic questions.

Acceptance Criteria

- Chatbot is available after login.
- Students receive relevant responses.
- Frequently asked questions are answered accurately.
- Chat history is available during the session.
- Unanswered questions are redirected to support.

User Story 08 (Priority: Medium)

As an Administrator,

I want AI to analyze attendance data,

So that abnormal attendance patterns can be detected automatically.

Acceptance Criteria

- AI analyzes attendance records.
- Abnormal patterns are highlighted.
- Reports are generated automatically.
- Admin receives alerts.
- Analysis can be exported.

User Story 09 (Priority: Medium)

As a Teacher,

I want AI to generate student performance reports,

So that I can evaluate student progress more efficiently.

Acceptance Criteria

- AI generates performance summaries.
- Reports include grades and attendance trends.
- Teachers can download reports as PDF.
- Reports are generated within one minute.
- Data accuracy is maintained.

User Story 10 (Priority: Low)

As an Administrator,

I want to manage the school website content,

So that announcements and information remain up to date.

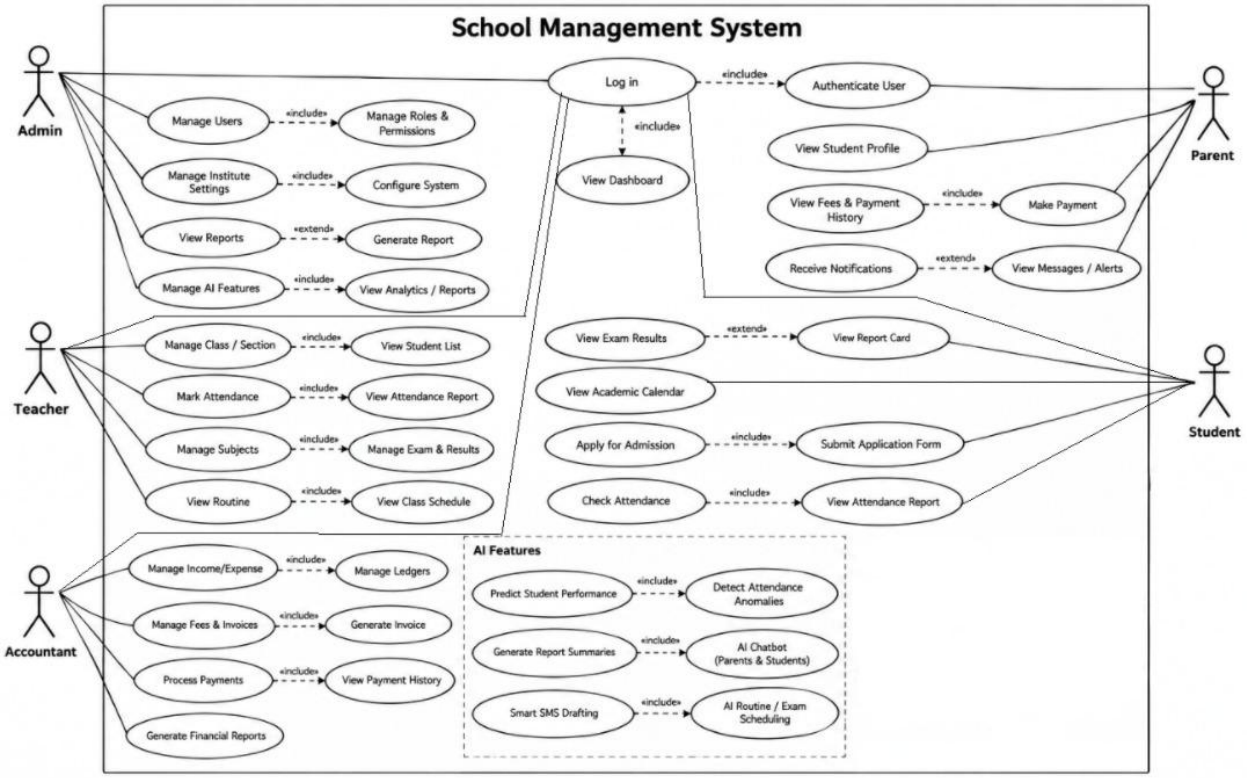
Acceptance Criteria

- Admin can create announcements.
- Admin can edit published content.
- Images and documents can be uploaded.
- Old notices can be archived.
- Changes are reflected immediately.

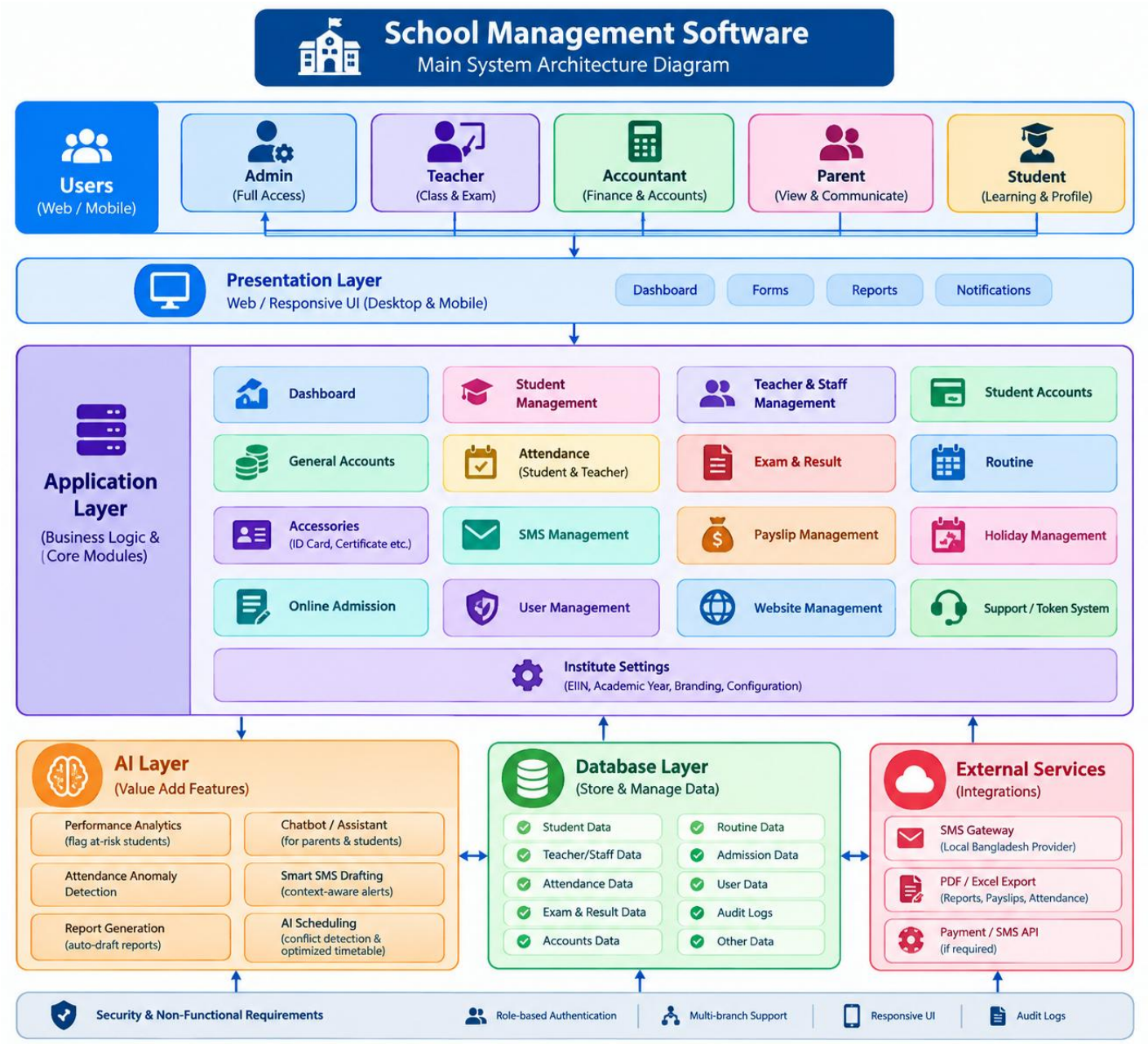
Priority Summary

User Story	Priority
US-01 Student Management	High
US-02 Attendance Management	High
US-03 Examination & Results	High
US-04 Fee Management	High
US-05 Parent Notification	Medium
US-06 Teacher & Staff Management	High
US-07 AI Chatbot	Medium
US-08 AI Attendance Analytics	Medium
US-09 AI Performance Report	Medium
US-10 Website CMS	Low

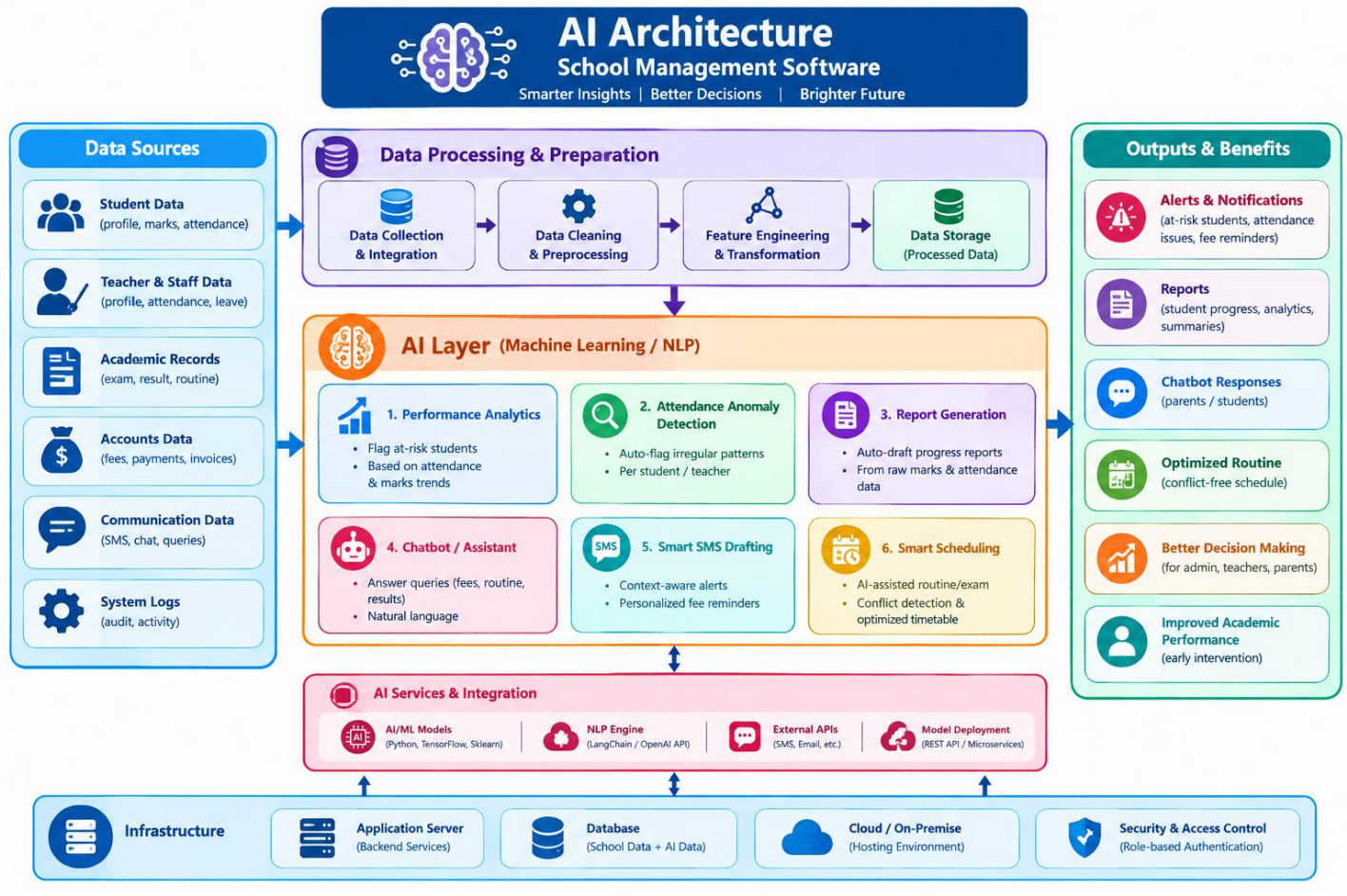
8. Use Case Diagram



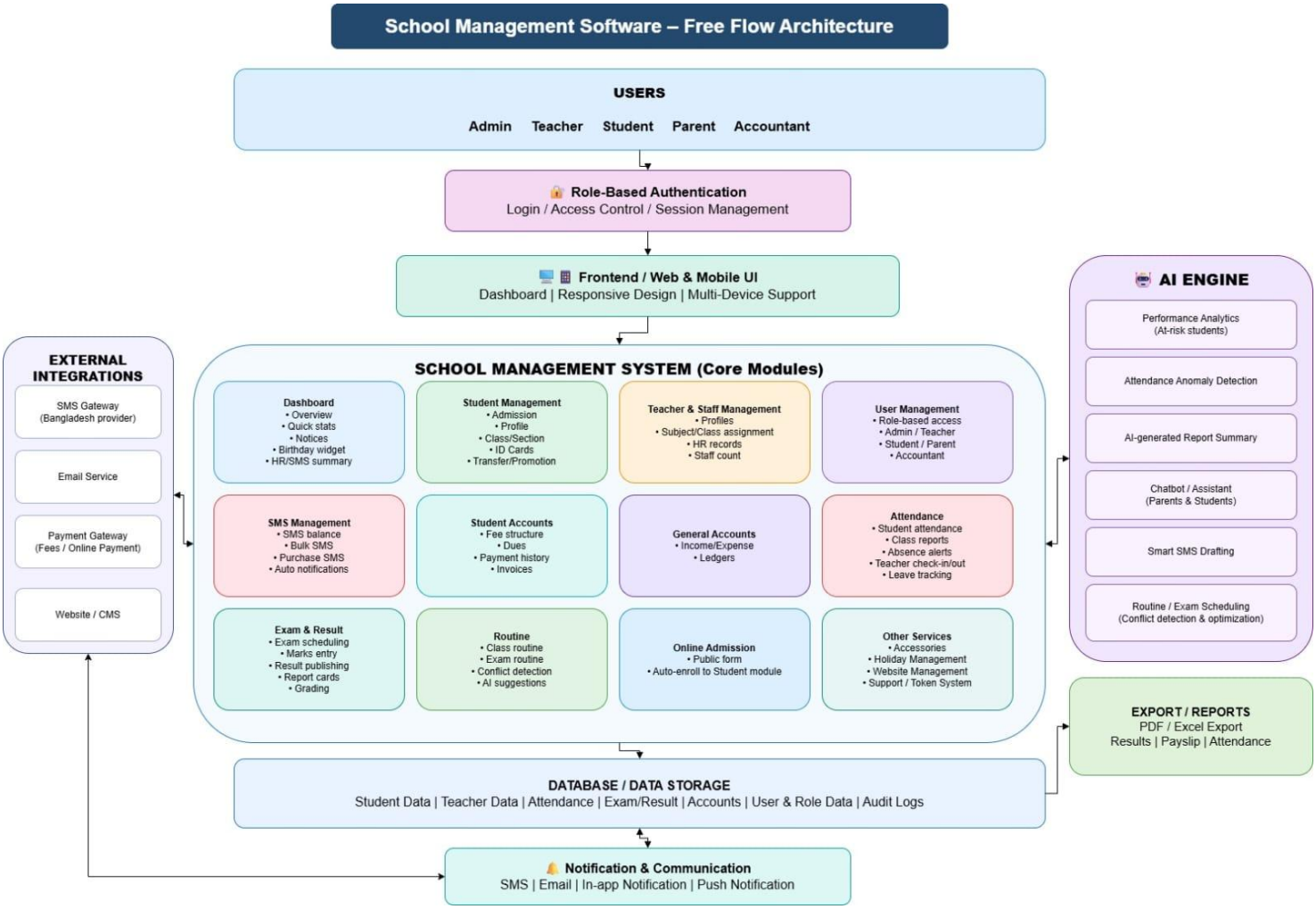
9. System Architecture



Picture: Main Architecture



Picture: AI Architecture



Picture: Free Flow Architecture

10. Artificial Intelligence Features

The proposed system integrates AI to improve decision-making and automation.

AI Features include:

- Student Performance Analytics
- Attendance Anomaly Detection
- AI Chatbot
- AI Report Generation
- Smart SMS Generation
- Timetable Optimization

11. Technology Stack

Frontend: React.js

Mobile: Flutter

Backend: Spring Boot

Database: PostgreSQL

Authentication: JWT

Artificial Intelligence: OpenAI / Gemini API

Deployment: Docker & AWS Cloud

12. Project Progress

Completed

- Requirement Gathering
- Client Interview
- User Stories
- Acceptance Criteria
- Use Case Diagram
- Software Architecture Design

In Progress

- UI/UX Design
- Database Design

Upcoming

- Backend Development
- Frontend Development
- AI Integration
- System Testing
- Deployment

13. Conclusion

The analysis and design phase of the School Management Software with AI Integration has been successfully completed. The project requirements have been gathered and documented, user stories have been prioritized, UML diagrams have been prepared, and the overall architecture has been designed. These deliverables provide a strong foundation for the implementation phase and ensure that the system will meet the client's functional and non-functional requirements.